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► **IN THE 20TH** century, oil emerged as the most critical commodity, reshaping geopolitics. Nations with vast oil reserves gained unprecedented economic and political influence. In the 21st century, rare earths are set to replace oil, driving the next wave of economic and geopolitical shifts. The reason: these 17 elements are indispensable to modern technologies such as smartphones, wind turbines, defence systems, and satellites. They are also the heart of the automotive industry as it moves towards electrification (they are used in high-performance magnets in EV motors and are crucial for its torque and energy efficiency).

However, as shortage of rare earth magnets due to China's export restrictions turns into a full-blown crisis, the global automotive industry, including domestic players, is looking at options. One of the things it has zeroed in on is ferrite magnets. Can these low-cost, non-conductive permanent magnets support the future of electric mobility?



“Rare earths are the new oil. We can’t electrify using globally constrained materials. That’s why we took a rare earth-agnostic approach”

ANKITH SOMANI
CO-FOUNDER, CONIFER.IO

SUBSTITUTING RARE EARTHS

In 2010, the global auto industry had faced a similar crisis when China cut its rare earth export quota by 40%, slashing exports from 50,000 tonnes in 2009 to about 30,259 tonnes. This drove up prices nearly sevenfold and ignited a global rush to develop rare earth mining and processing projects.

In April this year, the world faced a similar crisis when China imposed export restrictions in response to import tariffs by US President Donald Trump. The restrictions covered seven of the 17 elements—samarium, gadolinium, terbium, dysprosium, lutetium, scandium, and yttrium.

This has shaken up domestic players such as Bajaj Auto, Ather Energy, and TVS Motor, who anticipate that the shortage of rare earth magnets will hit EV production. Rating agency Crisil expects EV sales growth (expected 35-40%) to taper off. The panic was clearly visible in August. As per the Vahan data, Bajaj Auto slipped to the fifth position among electric two-wheeler manufacturers. Sales of the company's Chetak Electric Scooter and GoGo three-wheeler slumped to 8,940 units compared to 24,510 units in April.

China, however, eased some restrictions on India in August after its top officials met External Affairs Minister S. Jaishankar in Beijing. The situation, however, remains precarious, forcing automakers to pivot to alternatives. Some are looking at ferrite magnets as a long-term option. For example, two-wheeler manufacturer Ola Electric, in a letter to shareholders after the Q1FY26 results said the company has been working on using ferrite magnets for two years.

Mahindra & Mahindra, whose BE.6 and XEV.9E SUVs hit the market recently, has adopted a similar approach. Rajesh Jejurikar, ED & CEO-Auto & Farm, Mahindra & Mahindra, said at a recent press conference that the SUV manufacturer is comfortably covered for the next two quarters as it substituted rare earth with ferrite magnets. “There has been no disruption in production. We are covered for the coming quar-